SIAM DATA MINING 2026 · SALT LAKE CITY

# Sculpting the Latent Space

## Disentanglement with WAE-MMD and Programmable Priors

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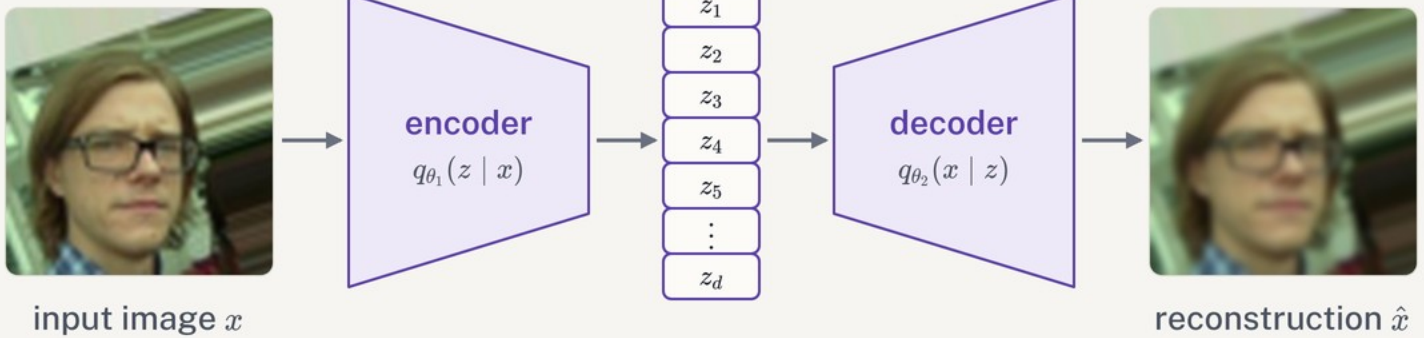
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THE SETTING

# An autoencoder compresses each image into a few numbers

the latent vector $z = (z_1, \ldots, z_d)$: $d$ numbers per image

($d = 32$ for CelebA faces)




input image $x$

reconstruction $\hat{x}$

trained end to end so that $\hat{x} \approx x$

**Encode every image: one dot each**




$$q_{\theta_1}(z) = \mathbb{E}_{x \sim p(x)}\left[q_{\theta_1}(z \mid x)\right]$$

the aggregate posterior: how the codes of all images are spread

CelebA face; the reconstruction is illustrative · encoder q(z | x) and decoder q(x | z) with weights $\theta_1$ and $\theta_2$, as in the paper

THE GOAL · DISENTANGLEMENT

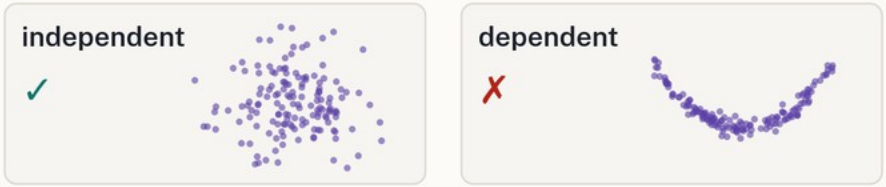
# Each latent should be independent and mean one thing




## (1) Independent

$$p(z) = \prod_{i=1}^{d} p(z_i)$$

Over the whole dataset, one latent tells you nothing about another: smiling says nothing about hair colour.




## (2) Interpretable

Each latent follows one real factor: slide it, and one thing changes.



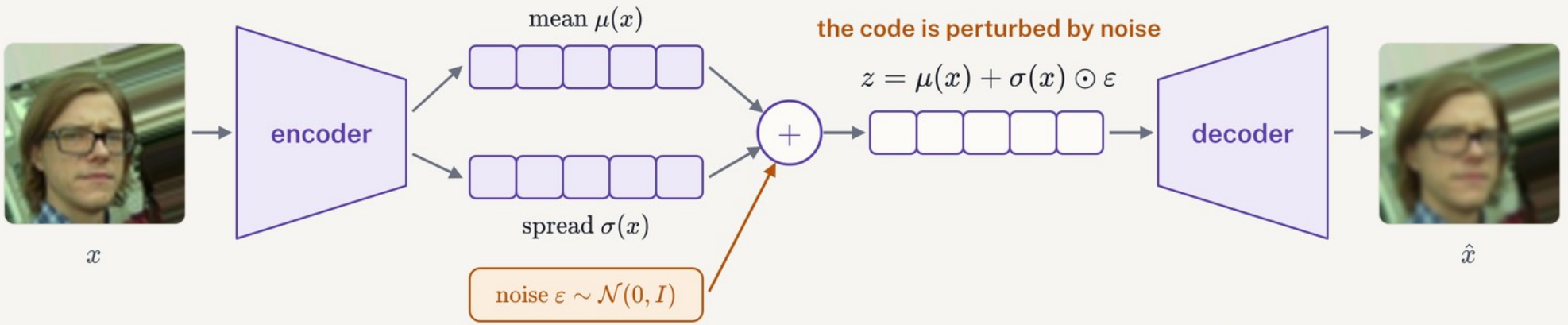

Our model on the XY dataset: one latent moves the circle horizontally, nothing else changes (paper, appendix).

**The catch:** without labels, (2) is not identifiable (nonlinear ICA): many independent representations fit the data equally well. So we measure and enforce (1), then add (2) with a few labels through the prior.

Attribute tags illustrative · Hyvärinen & Pajunen 1999; Locatello et al. 2019, 2020; Khemakhem et al. 2020

THE STANDARD RECIPE

# The variational autoencoder adds noise and a KL penalty

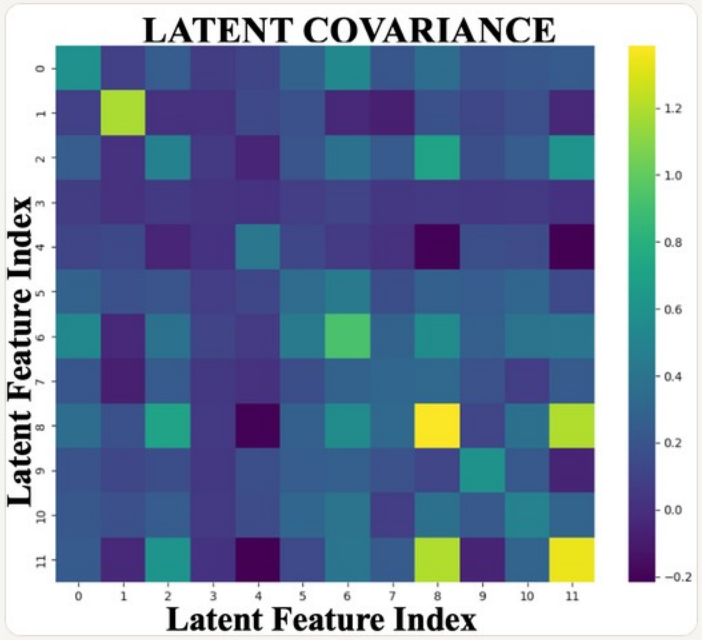



**The hope:** if every q(z | x) ≈ N(0, I), the aggregate q(z) = $E_x$[ q(z | x) ] ≈ N(0, I) too: a factorized prior, hence independent latents. β-VAE, FactorVAE and β-TCVAE all re-weight or split this KL term.

Kingma & Welling 2014 · Higgins et al. 2017 (β-VAE) · Kim & Mnih 2018 (FactorVAE) · Chen et al. 2018 (β-TCVAE)

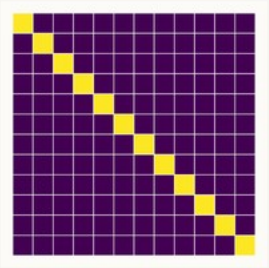
OUR MEASUREMENT · STANDARD VAE ON MNIST, D = 12

# Over the dataset, the codes are neither decorrelated nor Gaussian

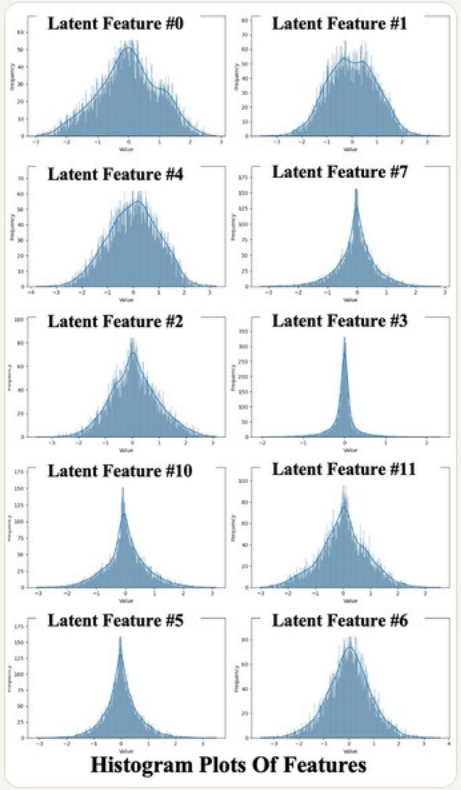



covariance of the 12 latents over all of MNIST

N(0, I) asks for:



the identity: no correlation




marginal of each latent

N(0, I) asks for:



a bell curve in every latent

If every q(z | x) were N(0, I), the aggregate would be too. It is not: the latents stay correlated, and their marginals are peaked and skewed.

We put a number on this dependence with LPS, a few slides on.

Paper, Figure 1(a): standard VAE trained on MNIST with 12 latents

WHY IT FAILS

# The KL term sees one code at a time, never the aggregate

### One input, one latent dimension

$$\mathrm{KL}\big(\mathcal{N}(\mu, \sigma^2) \,\|\, \mathcal{N}(0, 1)\big) = \tfrac{1}{2}\big(\mu^2 + \sigma^2 - 1 - \log \sigma^2\big)$$

Zero only at μ = 0, σ = 1, where the code forgets x. Reconstruction pulls the other way.

### All inputs together: the aggregate is a mixture

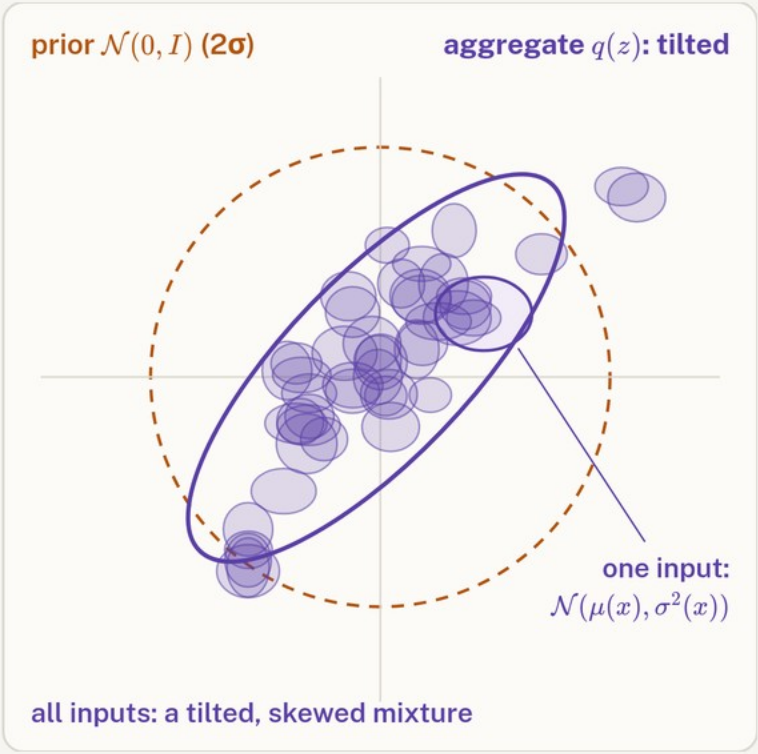
$$q(z) = \tfrac{1}{N} \textstyle\sum_{n=1}^{N} \mathcal{N}\big(\mu(x_n),\, \mathrm{diag}\, \sigma^2(x_n)\big)$$

$$\mathrm{Cov}_q[z] = \mathbb{E}_x\big[\mathrm{diag}\, \sigma^2(x)\big] + \mathrm{Cov}_x\big[\mu(x)\big]$$

### What the KL actually constrains

$$\mathbb{E}_x\big[\mathrm{KL}\big] = \textstyle\sum_{i=1}^{d} \tfrac{1}{2}\Big(\mathbb{E}\big[\mu_i^2\big] + \mathbb{E}\big[\sigma_i^2\big] - 1 - \mathbb{E}\big[\log \sigma_i^2\big]\Big)$$

**One dimension at a time: no term touches $\mathrm{Cov}_x(\mu_i, \mu_j)$ or the shape of the mixture.**

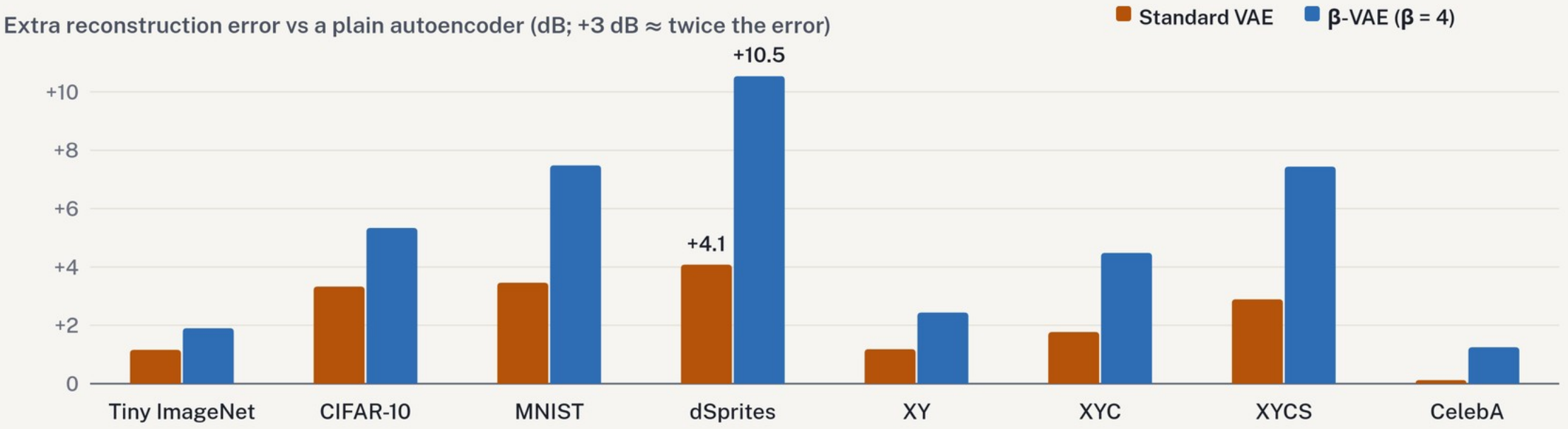



So the KL can only shrink the means (forget x) or inflate the spreads (drown x in noise). Neither makes the aggregate a factorized Gaussian, and nothing pushes its marginals to be Gaussian.

Right: schematic in two dimensions; ellipses are 2σ contours of q(z | $x_n$) and of the aggregate

THE OTHER COST · RECONSTRUCTION

# Noise in the code also degrades reconstruction




Disentangled codes are not only for generating new images: we also compress, edit and search images we already have, so reconstruction matters. On average the VAE costs +2.2 dB and β-VAE +5.1 dB. So we keep a deterministic autoencoder and regularize the aggregate instead.

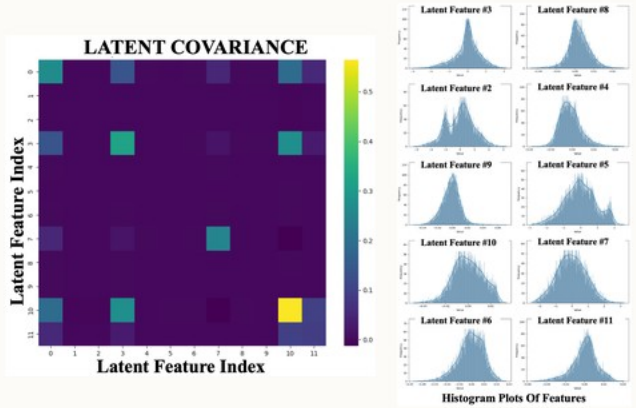
Reconstruction NMSE from the paper's Table 1, minus the standard AE's on the same dataset

TWO COMMON FIXES, AND OURS · MNIST, D = 12

# Only matching the whole aggregate distribution works

## β-VAE

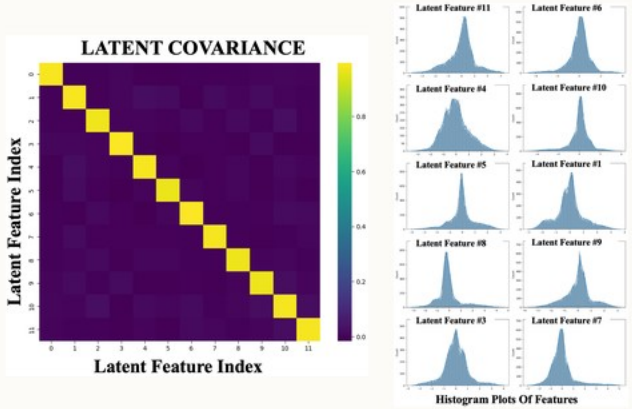
a stronger KL on each code (β = 4)




Still correlated; marginals still not Gaussian. **LPS 0.90**

## AE + batch KL

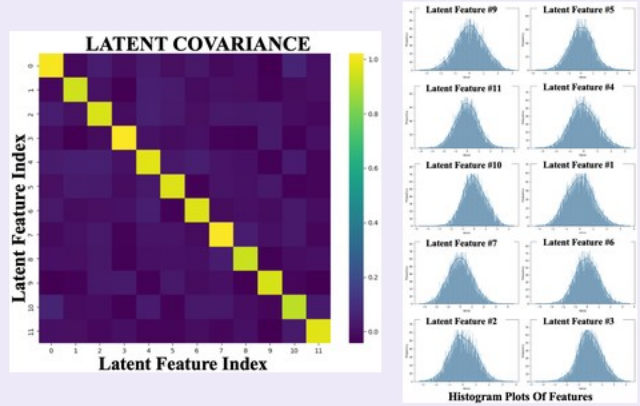
KL on the batch's mean and covariance




Covariance now diagonal, marginals far from Gaussian: still dependent. **LPS 0.97**

## SWAE-MMD (ours)

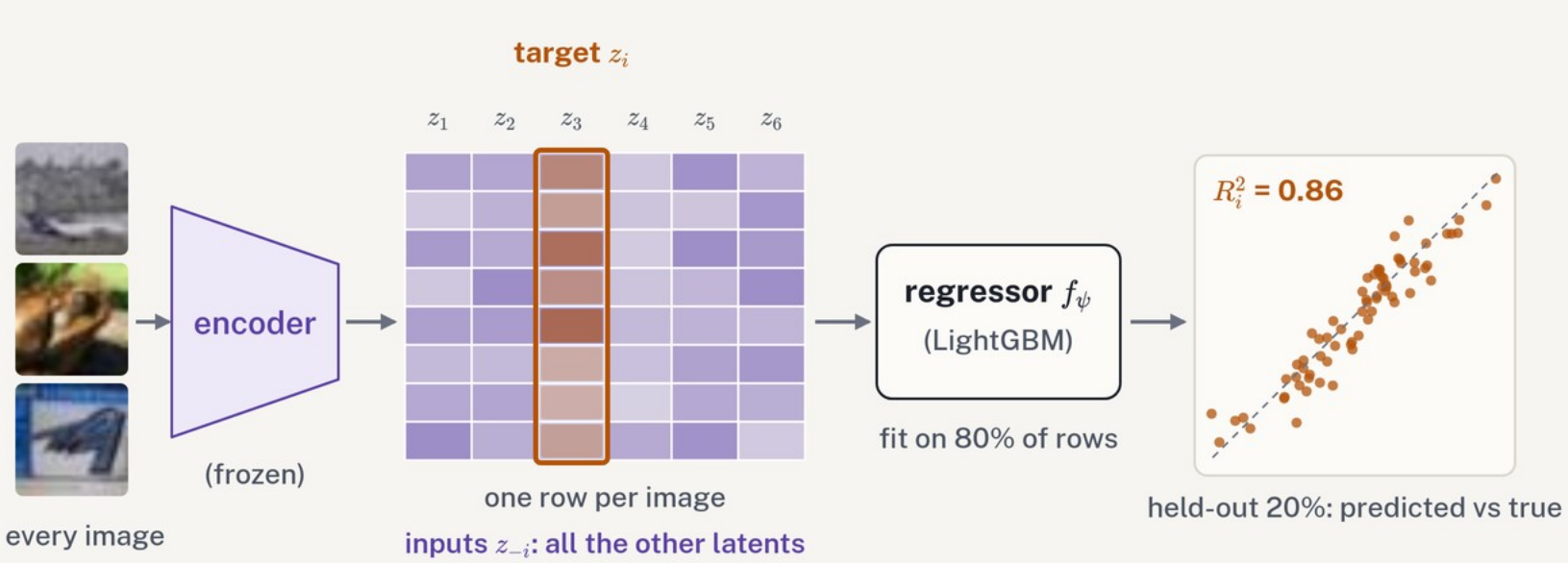
MMD between the batch and prior samples




Diagonal covariance and Gaussian marginals. **LPS 0.22**

Left: latent covariance over the dataset · right: the latents' marginals · LPS (lower = more independent) is defined next

# Can one latent be predicted from the others?




repeat for every latent, average:

$$\mathrm{LPS} = \frac{1}{d}\sum_{i=1}^{d} R_i^2$$




LPS from Table 1 (CIFAR-10); per-latent bars illustrative

LightGBM regressor, 80/20 split, up to 100,000 codes · LPS near 0: independent; near 1: fully predictable

WHY THE SCORE IS SAFE TO TRUST

# LPS can only underestimate entanglement

$R^2$ of any regressor $\leq \eta^2$, the share of a latent's variance the others truly explain

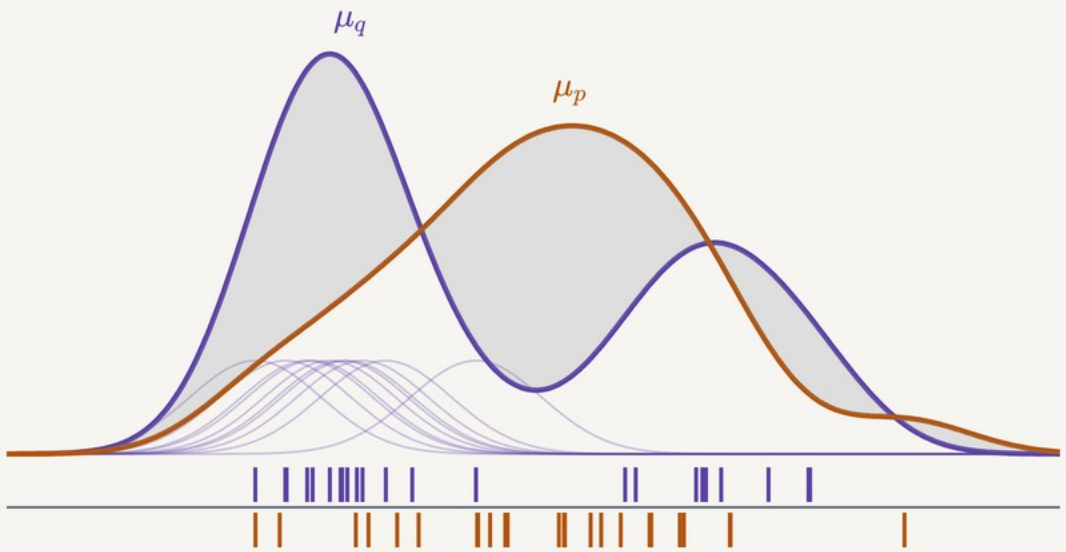
No regressor beats the conditional mean. So an LPS of 0.5 proves that at least half of the latent information is entangled, whatever the ground-truth factors are: a rejection test that needs no labels.

THE TOOL · MAXIMUM MEAN DISCREPANCY

# Comparing two distributions from samples alone

We want the aggregate q(z) to equal the prior p(z). But q(z) has no formula, only samples (a batch of codes), and the prior we can sample too. KL needs densities; a KL on batch statistics sees only a mean and a covariance.

a Gaussian bump on every sample, averaged: one curve per batch




encoded batch $z_i \sim q(z)$ (above) · prior batch $z'_j \sim p(z)$ (below)

**MMD = distance between the two curves**

$$\mathrm{MMD}(q,p) = \left\|\mu_q - \mu_p\right\|_{\mathcal{H}}, \qquad \mu_q = \mathbb{E}_{z\sim q}\left[k(z,\cdot)\right]$$

With a Gaussian kernel it is zero only if q = p: it compares whole distributions, not just a mean and a variance.

**From two batches, in closed form**

$$\widehat{\mathrm{MMD}}^2 = \tfrac{1}{m^2}\textstyle\sum_{i,j} k(z_i, z_j) + \tfrac{1}{n^2}\sum_{i,j} k(z'_i, z'_j) - \tfrac{2}{mn}\sum_{i,j} k(z_i, z'_j)$$

$$k(a,b) = \exp\!\left(-\|a-b\|^2/2\sigma^2\right)$$

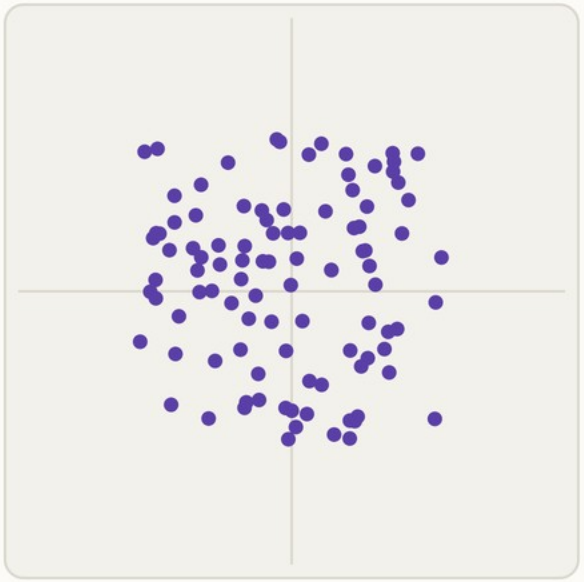
Differentiable in every code: its gradient moves the batch toward the prior. Bandwidths from the median heuristic.

Gretton et al., JMLR 2012 · shown in one dimension; the same estimator works on d-dimensional codes

SCULPTING, ONE BATCH AT A TIME

# MMD pulls the codes onto any prior

**loss = reconstruction error + λ · MMD²( latent batch, prior batch )**



**latent batch z = encoder(x)**
what the encoder produces

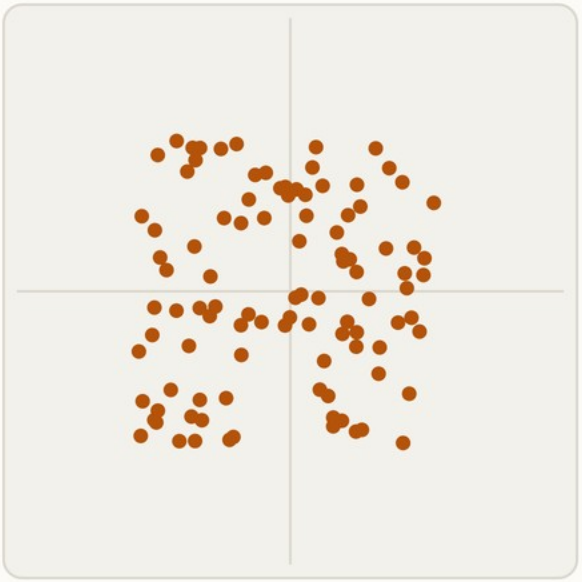
MMD²

**0.014**

correlation(z1, z2)

**-0.03**

**same loss, any prior you can sample from**



**prior batch z′ ~ p(z)**
prior: uniform

Two latent dimensions, schematic training · MMD² and the correlation are computed live from the points shown (Gaussian kernel)

OUR METHOD · SWAE-MMD

# WAE-MMD, extended to a programmable prior

WAE-MMD · Tolstikhin et al., ICLR 2018

$$\mathcal{L}_{\text{WAE}} = \mathbb{E}_{p(x)}\big[\log q_\theta(x)\big] - \lambda\,\text{MMD}^2\big(q_{\theta_1}(z),\, p(z)\big)$$

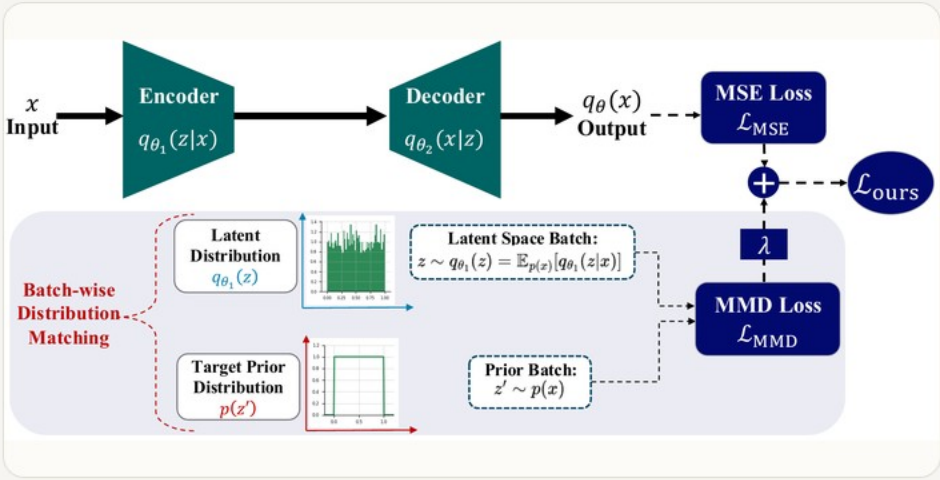
$p(z) = \mathcal{N}(0, I)$, built to generate images

**SWAE-MMD · ours**

$$\mathcal{L}_{\text{ours}} = \mathbb{E}_{p(x)}\big[\log q_\theta(x)\big] - \lambda\,\text{MMD}^2\big(q_{\theta_1}(z, f_\phi(u)),\, p(z, f_\phi(u))\big)$$

- any prior $p$ we can sample: factorized → independence; its shape → compression
- joint with pseudo-labels $f_\phi(u)$ → alignment to chosen factors

No labels and $p = \mathcal{N}(0, I)$: exactly WAE-MMD. Goal: representations, not generation.

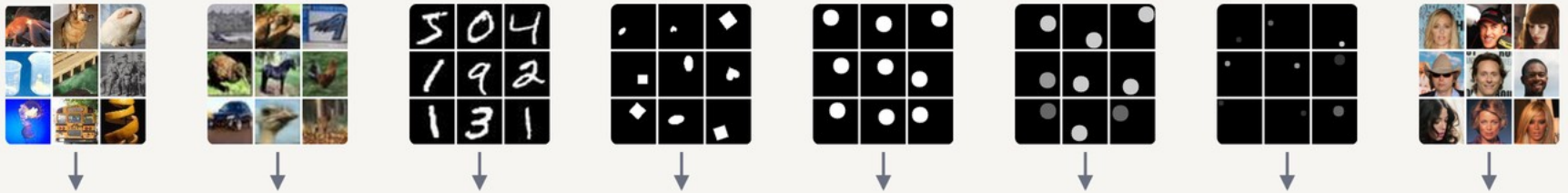



A batch of codes and a batch sampled from the prior go into the MMD; the decoder keeps its reconstruction loss. The regularizer sees the whole batch, so it shapes the aggregate.

RESULTS · 8 DATASETS

# 70% less entanglement than the next best method

a sample of each dataset



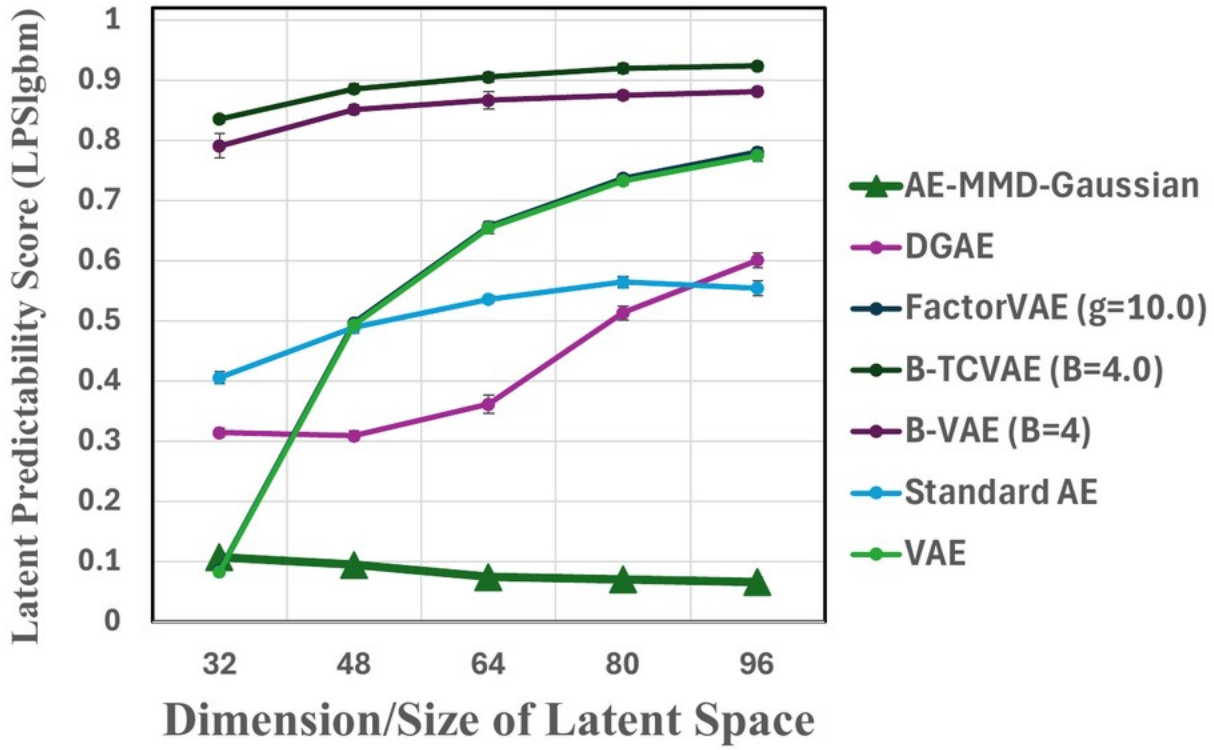
| model | ImageNet [64] | CIFAR-10 [64] | MNIST [12] | dSprites [5] | XY [2] | XYC [3] | XYCS [4] | CelebA [32] |
|---|---|---|---|---|---|---|---|---|
| **SWAE-MMD (ours)** | **0.03** | **0.04** | **0.22** | **0.04** | −0.05 | **−0.01** | **0.11** | 0.08 |
| FactorVAE (γ = 10) | 0.44 | 0.63 | 0.27 | 0.08 | −0.04 | 0.00 | 0.17 | **0.07** |
| NashAE | 0.56 | 0.17 | **0.22** | 0.10 | **−0.07** | 0.14 | 0.37 | 0.36 |
| β-TCVAE (β = 4) | 0.79 | 0.89 | 0.84 | 0.22 | −0.04 | **−0.01** | 0.17 | 0.59 |
| β-VAE (β = 4) | 0.73 | 0.85 | 0.90 | 0.52 | −0.05 | 0.00 | 0.28 | 0.53 |
| DGAE | 0.77 | 0.69 | 0.63 | 0.56 | 0.25 | 0.50 | 0.59 | 0.48 |

Averaged over the 8 datasets, LPS falls by 70% against FactorVAE, with reconstruction as good or better (CIFAR-10: −15.61 dB NMSE vs −12.49 dB).

LPS, lower = more independent · latent size in brackets · ImageNet = Tiny ImageNet · Table 1 of the paper · bold = best in column

RESULTS · CIFAR-10

# Independence holds as the latent space grows




## Flat in the latent size

From 32 to 96 dimensions the MMD model stays low, at about 0.1 or below, while most baselines grow more entangled.

## Insensitive to its knobs

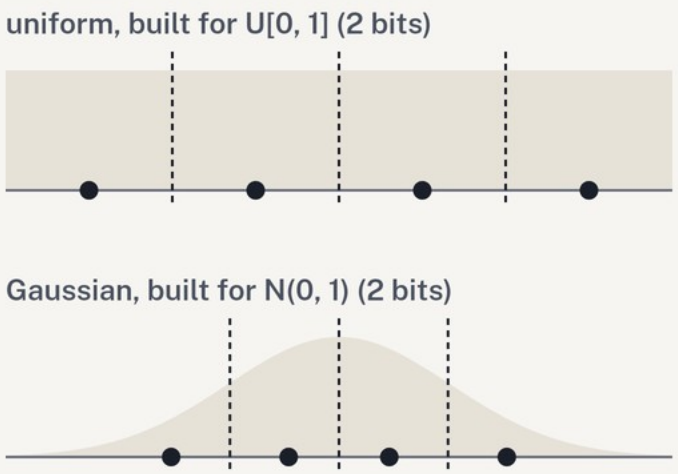
Batch 64 to 2048, λ from 0.1 to 5, 1 to 5 kernel bandwidths: only small changes in LPS and reconstruction.

LPS with a LightGBM regressor; lower is better · ablations in the paper's appendix

PROGRAMMABLE PRIOR · COMPRESSION

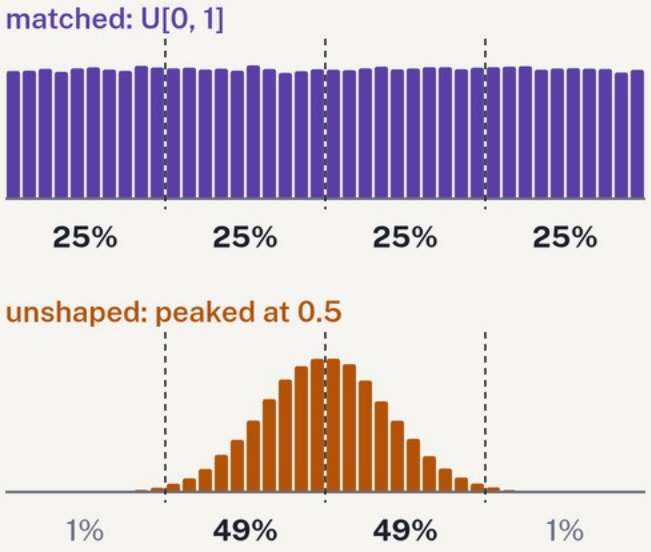
# A quantizer only works if the codes have the shape it expects

**A $b$-bit quantizer snaps each latent to one of $2^b$ levels**



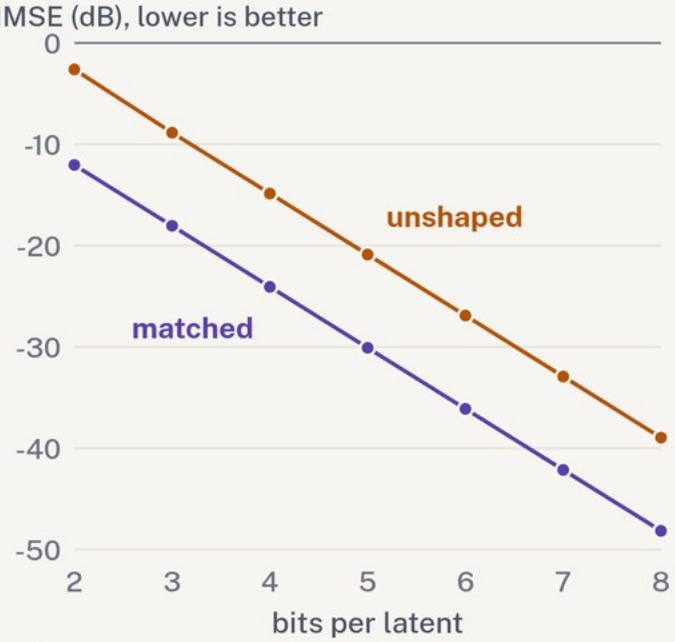

dots: the levels · dashed: bin edges · grey: the density the quantizer assumes

**Same 2-bit uniform quantizer, two latents**




share of codes per level: the peaked latent uses 2 of 4 levels

**Quantization error vs bits (simulated)**




So we set the prior to the quantizer's distribution and train with the quantizer in the loop: MMD shapes every latent to use all of the levels.

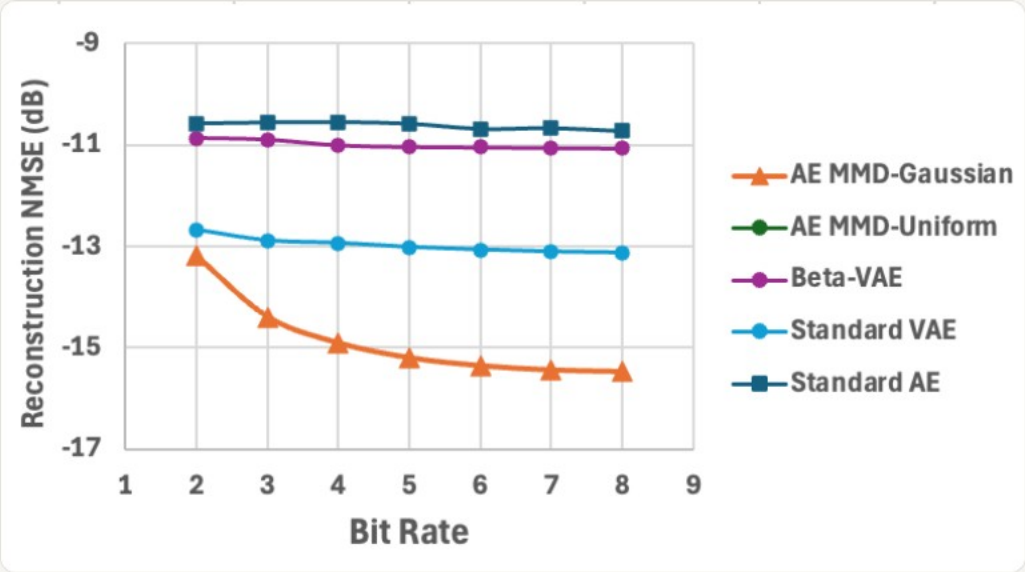
Illustrative 2-bit designs (uniform on [0, 1]; Lloyd–Max for N(0, 1)) · simulated latents, not a trained model

COMPRESSION RESULTS · CIFAR-10

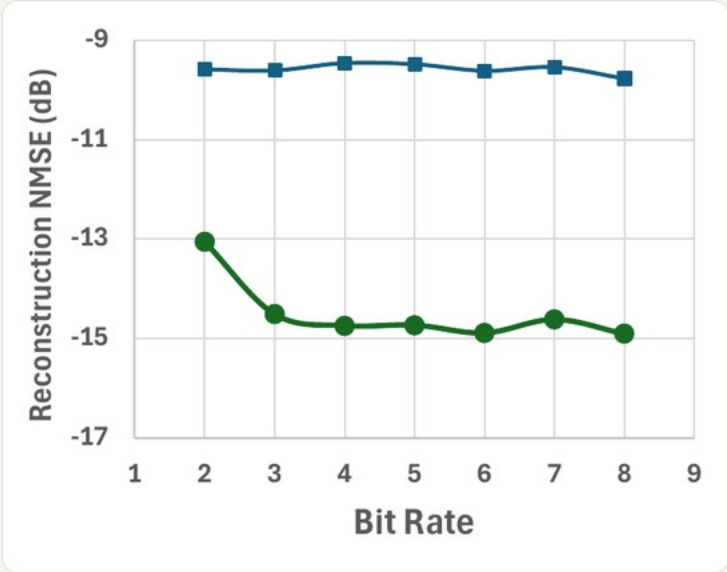
# Shape the codes to fit the quantizer

Train with the quantizer in the loop and a prior equal to the distribution the quantizer assumes, Gaussian or uniform.

**Gaussian quantizer · CIFAR-10**




**Uniform quantizer · CIFAR-10**



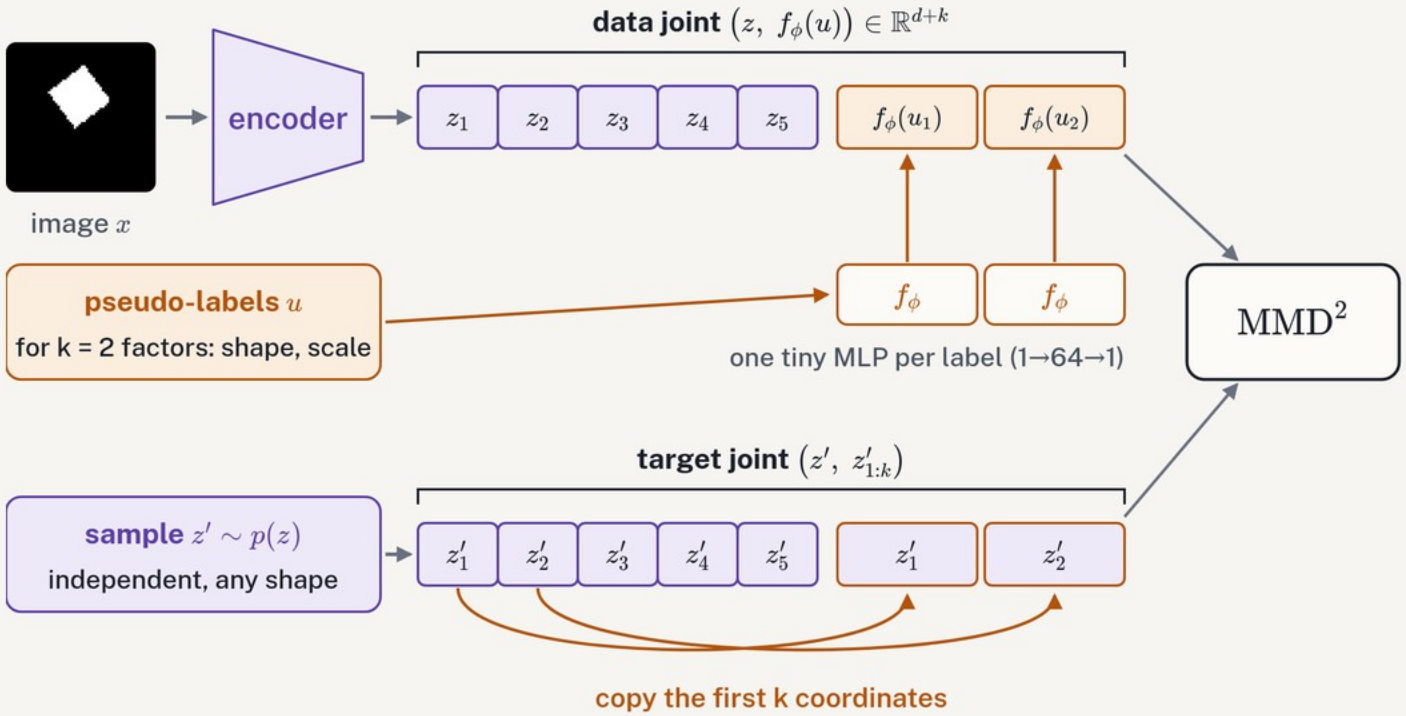

**2–7 dB lower reconstruction error** at 2 to 8 bits per dimension, with no post-training vector quantization or entropy coding.

Reconstruction NMSE in dB, lower is better · the MMD model matched to the quantizer: orange (Gaussian), green (uniform)

PROGRAMMABLE PRIOR · ALIGNMENT

# Pin chosen latents to a few labels through the prior

Independent latents are still arbitrary: nothing says which one is shape. A few labelled images fix that.




**To match the target, the data side must copy too:**

$$z_1 = f_\phi(u_1), \quad z_2 = f_\phi(u_2)$$

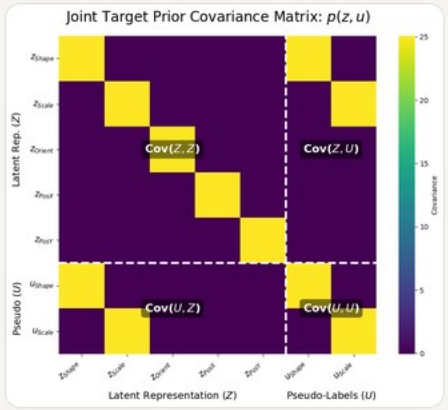
latent 1 now carries shape, latent 2 carries scale, and $z$ still follows $p(z)$: independent and shaped.

$f_\phi$ only rescales each label onto the prior's marginal. Images without labels use $(z,\ z_{1:k})$ and satisfy the copy for free.

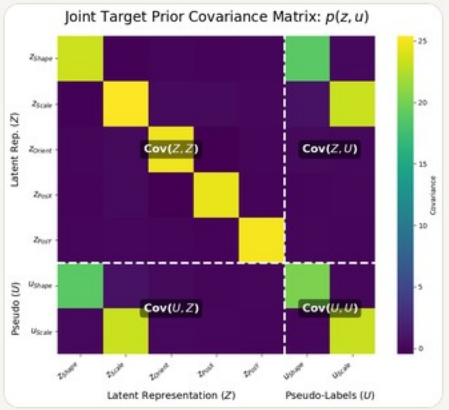
dSprites: 5 factors (shape, scale, orientation, x, y); k = 2 labelled here · label maps: 1 → 64 → 1 MLPs with BatchNorm

ALIGNMENT RESULTS · DSPRITES

# A few pseudo-labels lock latents to real factors

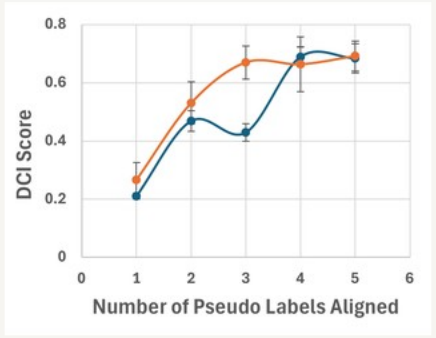



target

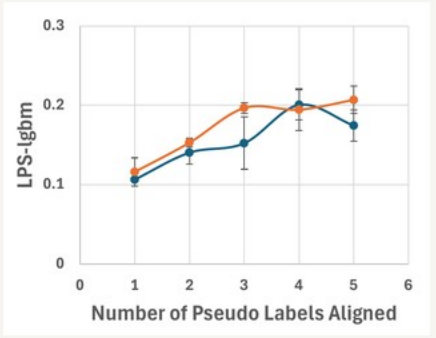



learned

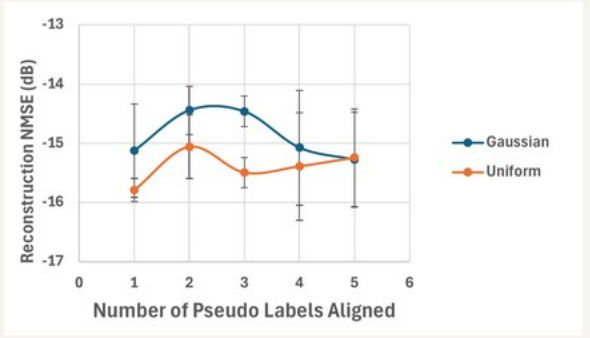
Each covariance compares the latents z with the labels f(u). In the Cov(Z, U) block the two labels line up one-to-one with latents 1 and 2, as the copied target demands, and Cov(Z, Z) stays diagonal: the latents are still independent.




DCI disentanglement ↑




LPS ↓




reconstruction NMSE (dB) ↓

DCI rises sharply, peaking at 3 of 5 factors (uniform prior) and 4 of 5 (Gaussian). Reported weakly supervised baselines (Ada-GVAE, Ada-ML-VAE) stay below 0.5.

LPS rises too: dSprites' own factors are not independent, so (1) and (2) conflict here. Reconstruction stays flat.

x-axes: pseudo-labels aligned (of 5) · blue: Gaussian prior, orange: uniform prior · DCI always against all 5 factors

TAKEAWAYS

# The prior is an engineering tool, not a fixed assumption

## A label-free test

LPS lower-bounds entanglement and exposes the VAE-style failure.

## A regularizer that works

MMD on the aggregate posterior: 70% less entanglement, reconstruction kept.

## One loss, many priors

Semantic alignment, quantizer matching and latent copying.

JOINT WORK WITH




**Akshay Malhotra**
InterDigital



**Shahab Hamidi-Rad**
InterDigital



**Aditya Sant**
InterDigital



**Aryan Mokhtari**
UT Austin



**Sujay Sanghavi**
UT Austin

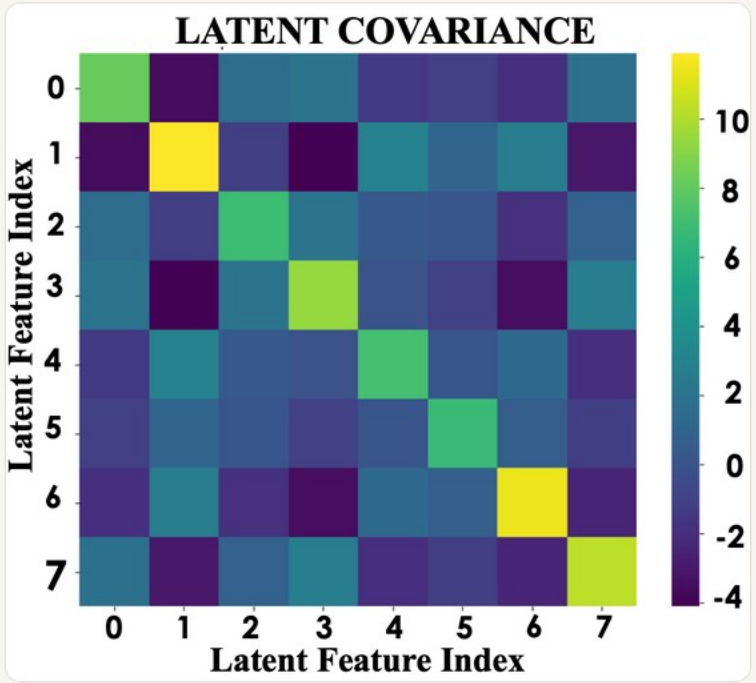

arxiv.org/abs/2510.11953 · github.com/Quentin-Fruytier/Sculpting-Latent-Spaces-With-MMD

Work done at InterDigital AI Lab with the University of Texas at Austin · thank you

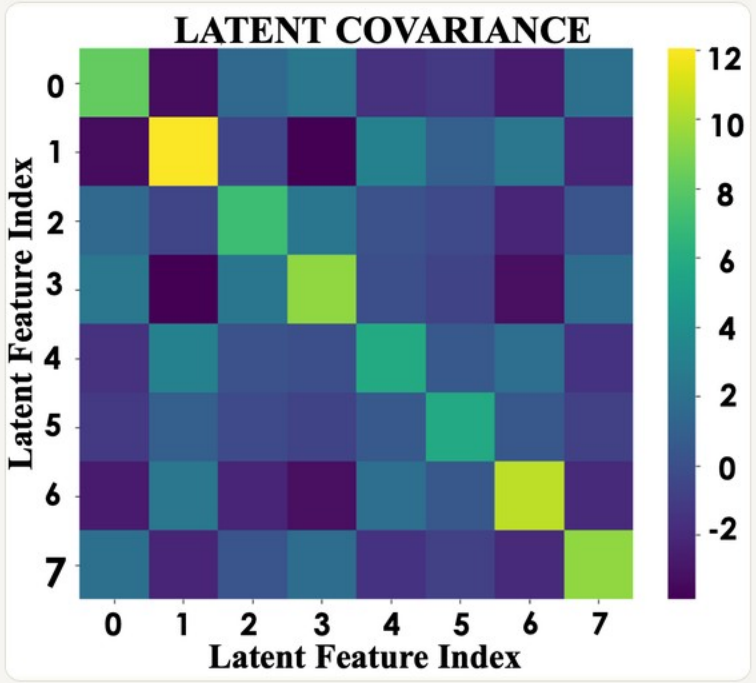
BACKUP · A PRIOR IS ANYTHING YOU CAN SAMPLE

# MMD can copy another model's latent space

teacher: a standard AE's codes, used as the prior




student: trained with MMD to match them




An entangled, co-dependent geometry that no closed-form KL target could express.

Covariance and marginals come out nearly identical.

MNIST, latent covariance over the dataset · paper appendix, latent copying experiment